

A Total Energy Consumption Model

In energy-harvesting or battery-free terminal devices, the available energy supply is not only highly constrained but also exhibits strong temporal intermittency due to dependence on environmental energy sources such as RF power, solar irradiance, or ambient vibrations. These fluctuations create a fundamentally different operational regime compared to battery-powered designs: instead of assuming continuous power availability, systems must constantly adjust their execution strategies to match the instantaneous energy budget. Therefore, a precise and analytically tractable energy consumption model is crucial for enabling adaptive scheduling, power-aware security, operation scaling, and safe task completion.

From a system-level perspective, the total energy dissipation of a battery-free node can be classified into two major functional domains. The first is the communication-related energy E_{comm} , which includes all RF transmission, reception, and protocol overhead. The second is the on-device computation energy E_{local} , which encompasses processor switching activity, leakage currents, peripheral usage, and memory hierarchy operations. These two categories together form the total consumption:

$$E_{\text{total}} = E_{\text{comm}} + E_{\text{local}}. \quad (1)$$

This decomposition is not merely conceptual but also reflects distinct controllability characteristics. Communication energy is largely influenced by environmental factors such as channel fading, interference, distance, and antenna orientation—many of which are outside the system’s direct control. In contrast, local computation energy is more structurally deterministic and can be optimized through DVFS, clock gating, task partitioning, or algorithm redesign. Balancing these two domains is central to designing robust energy-adaptive systems capable of maintaining functionality under unpredictable power availability. When communication cost dominates, the node benefits from computation-heavy local processing; conversely, when local computation is energy-intensive, task offloading may become advantageous. The unified model thus provides the foundation for cross-layer optimization in energy-constrained IoT systems.

A.1 Communication Energy Consumption Modeling

Communication typically constitutes the largest share of energy consumption in battery-free nodes (BFNs). Due to the low efficiency of micro-scale antennas and limited rectifier conversion efficiency, the radio subsystem often consumes more energy than the computation unit by one to two orders of magnitude. The total communication energy is decomposed according to:

$$E_{\text{comm}} = E_{\text{tx}} + E_{\text{rx}} + E_{\text{ctrl}} + E_{\text{sec}},$$

where each term captures a distinct category of communication operations. This decomposition enables fine-grained optimization, as each term responds differ-

ently to environmental conditions, protocol states, modulation configurations, and security requirements.

A.1.1 Transmission Energy Consumption

Transmission is the most energy-demanding operation because generating a stable RF signal at sufficient power to overcome channel attenuation requires substantial circuit and amplifier effort. The energy for transmitting N packets of payload size L is given by

$$E_{\text{tx}} = N \times \left((P_c + \eta \cdot P_{\text{te}}) \times \frac{L}{R} \right), \quad (2)$$

where the transmission duration $\frac{L}{R}$ encapsulates the impact of physical-layer rate R on temporal energy cost.

The circuit power P_c contains a static baseline component P_{base} and a modulation-related dynamic component P_{mod} :

$$P_c = P_{\text{base}} + P_{\text{mod}}, \quad P_{\text{mod}} \propto R^\beta, \quad \beta \approx 1. \quad (3)$$

Higher transmission rates require more sophisticated modulation and coding, increasing oscillator activity, mixer load, and filter switching, all of which contribute to elevated dynamic circuit power.

To maintain a target SNR at the receiver, the required transmission power must compensate for propagation loss:

$$P_{\text{te}} = \frac{\text{SNR} \cdot N_0 \cdot d^\alpha}{\eta \cdot h}, \quad (4)$$

with d and α controlling the geometric decay of signal power and h representing random fading. This relationship clearly shows that environmental conditions heavily influence energy cost, and poor channel gain quickly renders transmission unaffordable under limited harvested energy.

Power amplifier efficiency follows a saturation response:

$$\eta = \eta_{\text{max}} \cdot (1 - e^{-\gamma P_{\text{te}}}), \quad (5)$$

indicating diminishing efficiency returns as transmission power increases. For BFNs, this non-linearity greatly affects optimal packet size and modulation choices.

The payload size and data rate are computed as:

$$L = \left(\left\lceil \frac{\text{size}}{\text{MTU}} \right\rceil \cdot B + \text{size} \right) \cdot 8, \quad (6)$$

$$R = B \cdot \log_2 \left(1 + \frac{S}{N + N_{\text{interf}}} \right) \cdot C, \quad (7)$$

reflecting the impact of interference, modulation fidelity, and hardware capability on communication efficiency. This formulation emphasizes that link quality directly propagates into energy cost.

A.1.2 Reception Energy Consumption

Although reception is less energy-intensive than transmission, it remains a significant contributor due to extended listening periods, synchronization tracking, and preamble decoding. The reception energy model is

$$E_{\text{rx}} = N \cdot \left(P_{\text{rx}} \cdot \frac{L}{R} \right) + P_{\text{idle}} \cdot T_{\text{idle}}. \quad (8)$$

Battery-free nodes often cannot afford long idle-listening durations. Low-Power Listening (LPL) therefore replaces continuous reception with periodic channel sampling:

$$E_{\text{rx}}^{\text{LPL}} = N \cdot \left(P_{\text{rx}} \frac{L}{R} \right) + (P_{\text{wake}} T_{\text{wake}} + P_{\text{sleep}} T_{\text{sleep}}) \frac{T_{\text{idle}}}{T_{\text{cycle}}}. \quad (9)$$

Shorter sleep cycles reduce communication delay but substantially increase energy consumption. As a result, LPL cycle length is a critical parameter determining system responsiveness.

A.1.3 Protocol Control Overhead

Control signaling includes handshake packets, channel probing, ACK/NACK exchange, and synchronization preambles, all of which incur extra cost beyond payload transmission:

$$E_{\text{ctrl}} = N_{\text{ctrl}} \left(\frac{P_{\text{tx}} C}{R} + \frac{P_{\text{rx}} C}{R} \right) + E_{\text{proc}} N. \quad (10)$$

In LPL-enabled nodes, additional wake-ups multiply this overhead:

$$E_{\text{ctrl}}^{\text{LPL}} = N_{\text{wake}} \left[N_{\text{ctrl}}^{\text{per}} \frac{(P_{\text{tx}} + P_{\text{rx}}) C}{R} + E_{\text{proc}} N_{\text{ctrl}}^{\text{per}} \right]. \quad (11)$$

This cost becomes especially relevant when the channel is noisy or unstable, as frequent retransmissions amplify control overhead exponentially.

A.1.4 Secure Transmission Overhead

Security operations introduce substantial computational energy due to cryptographic transformations such as permutations, modular additions, hashing, or MAC generation:

$$E_{\text{sec}} = N \cdot (E_{\text{enc}} \cdot L + E_{\text{dec}} \cdot L + E_{\text{auth}}). \quad (12)$$

If the system supports configurable security levels l , the generalized model becomes

$$E_{\text{sec}} = N \cdot (E_{\text{enc}}(l) L + E_{\text{dec}}(l) L + E_{\text{auth}}(l)). \quad (13)$$

Security energy is therefore directly tunable, enabling adaptive security that matches energy availability.

A.2 Modeling of Local Processing Energy Consumption

Local energy consumption originates from processor switching activities, memory accesses, leakage currents, and operation of peripheral units such as timers, ADCs, and communication controllers. Unlike communication, local computation is more predictable and thus easier to optimize. Figure ?? highlights the qualitative behavior across different operation modes.

A.2.1 System Load Power Consumption

The processor's power consumption consists of three major components:

$$P_{\text{dyn,sys}}(f) = \alpha C_{\text{eq}} V_{\text{op}}^2 f, \quad (14)$$

$$P_{\text{ind}} = P_{\text{mem}} + P_{\text{IO}} + P_{\text{aux}}, \quad (15)$$

$$P_{\text{sta}} = V \cdot I_{\text{leak}}. \quad (16)$$

Dynamic power grows linearly with frequency but quadratically with voltage, which makes DVFS particularly effective when voltage can be reduced substantially. Static leakage becomes dominant at low voltage and long execution times, which is common in energy-harvesting systems where computation may pause and resume repeatedly. Peripheral power P_{ind} becomes important in sensor-heavy workloads, where long sampling intervals may exceed CPU active energy.

A.2.2 Task Execution Time

Task duration directly affects total energy through the standard relation $E = P \cdot T$. For parallelizable operations:

$$T = \max \left(\frac{C}{f_{\text{cpu}}}, \frac{D_{\text{tx}} + D_{\text{rx}}}{B}, \frac{S}{R_{\text{sto}}} \right), \quad (17)$$

while serial operations satisfy:

$$T = \frac{C}{f_{\text{cpu}}} + \frac{D_{\text{tx}} + D_{\text{rx}}}{B} + \frac{S}{R_{\text{sto}}}. \quad (18)$$

These expressions demonstrate that task completion depends not only on CPU speed but also on communication and storage throughput. Under intermittent energy supply, these equations help predict whether a task segment can finish before power loss, enabling proactive checkpointing or task migration.

Summary

This appendix provides supplementary details of the proposed energy-driven task scheduling mechanism, including implementation examples, data analysis, and experimental results. The additional figures and explanations are intended

to complement the main body of the paper by clarifying design choices, demonstrating performance trade-offs, and validating the effectiveness of the approach under dynamic energy conditions. Together, these results further confirm the practicality and adaptability of the proposed mechanism in energy-constrained IoT environments.